Persistence Without Contradiction: A Minimal Foundation for Existence

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Abstract

A form exists as a determinate entity if and only if it satisfies two independent structural conditions: idempotent closure—stable self-identity under re-application of its defining boundary—and energetic viability—maintenance cost within available sustaining capacity. Forms violating either condition do not exist as determinate entities. The constraint is eliminative, not probabilistic, and applies to itself without regress.[1]

1 The Constraint

The argument rests on three primitives. A state space S is a nonempty set of candidates. A form is any $\psi \in S$ together with an identity condition $\equiv_\psi$ specifying which states count as the same form across transformations. The admissible transformation family $\Delta(\psi)$ is the nonempty set of maps $\delta : S \to S$ that are constitutive of ψ's claimed structural role, closed under composition, and fixed prior to evaluation—not chosen post hoc to preserve ψ.[2]

With these primitives in place, the central definition is as follows.

Definition 1.1 (Determinate Existence)**.** *A form $\psi \in S$ exists as a determinate entity if and only if:*

(i) There exists a selective criterion $C_\psi : S \to \{0, 1\}$ with $C_\psi(\psi) = 1$ and $\exists \phi \in S$ with $C_\psi(\phi) = 0$.

(ii) C_ψ is stable under $\Delta(\psi)$: for all $\delta \in \Delta(\psi)$, $C_\psi(\delta(\psi)) = 1$.

[1] A complete Lean 4 formalization of all results in this paper is available at DOI: 10.5281/zenodo.19023530.

[2] The requirement that transformations be fixed prior to evaluation is not an epistemic restriction; it is a structural one. If transformations could be chosen post hoc, any form could be made to seem persistent by discarding disconfirming transformations. The constraint would be trivial.

(iii) The maintenance cost functional $W_\psi : \Delta(\psi) \times \mathbb{N} \to \mathbb{R}_{\geq 0}$ satisfies energetic viability (Definition 2.2).

If any condition fails, the expression 'ψ exists' lacks determinate referent.

2 The Two Conditions

The first condition concerns consistency of identity under transformation.

Definition 2.1 (Idempotent Closure)**.** *A form ψ satisfies idempotent closure if for all $n \in \mathbb{N}$ and all $\delta \in \Delta(\psi)$:*

$$\psi \equiv_\psi \delta^n(\psi).$$

The form returns the same form under iterated re-application of its defining boundary. Drift and oscillation are excluded.

The second condition concerns the resource cost of maintaining that consistency over time.

Definition 2.2 (Energetic Viability)**.** *Let $W_\psi(\delta, n)$ denote the cost of maintaining ψ's boundary through n successive admissible updates under δ, and let $\Phi_\psi(n)$ denote available sustaining capacity at step n. A form ψ satisfies energetic viability if:*

$$\forall n \in \mathbb{N}, \forall \delta \in \Delta(\psi) : \Phi_\psi(n) \geq W_\psi(\delta, n).$$

A form whose maintenance cost is unbounded relative to any finite sustaining capacity fails this condition.[3]

A form may satisfy one condition and fail the other. The conditions filter independently.

Theorem 2.3 (Independence)**.** *There exists a form satisfying idempotent closure that fails energetic viability.*

Proof. Define ψ by the identity condition: maintain a complete lossless record of all prior states indefinitely. The rule is internally consistent—idempotent closure holds. The maintenance cost $W_\psi(\delta, n)$ grows strictly with n, exceeding any finite $\Phi_{\max}(n) < \infty$ for sufficiently large n. Energetic viability therefore fails. □

A third concept is required to state the eliminative consequence.

Definition 2.4 (Globally Coupled Contradiction)**.** *A form ψ contains a globally coupled contradiction (GCC) if there exists a predicate P such that ψ's identity conditions require both $P(\psi)$ and $\neg P(\psi)$, and no stable witness structure π separates P from $\neg P$—by localization, indexing, staging, or confinement—while preserving ψ under $\Delta(\psi)$.*

[3] The terms *cost* and *capacity* are not restricted to physical energy. They refer to any finite resource required to preserve a distinction: memory, representational effort, proof length, etc. The substrate determines the metric.

3 Energetic Viability as a Structural Constraint

A persistent concern must be addressed directly: is energetic viability merely a physicalist assumption smuggled into a structural ontology? The independence theorem demonstrates it is not. But independence alone does not establish positive content. This section provides that content.

Definition 3.1 (Cost-Bearing Substrate)**.** *For any form ψ, let $\Sigma(\psi)$ denote the substrate on which ψ is realized—the medium, domain, or background structure that supports the instantiation of ψ's boundary conditions and carries the transformations in $\Delta(\psi)$. The substrate is not assumed to be physical. It may be:*

- *a representational medium (language, formal system, memory);*
- *a computational architecture (Turing machine, neural network);*
- *a mathematical domain (set-theoretic universe, type theory);*
- *a phenomenal field (conscious experience, qualia space);*
- *any structure capable of supporting re-identifiable forms.*

The only assumption about $\Sigma(\psi)$ is that it has finite local capacity*: at any given stage of evaluation or transformation, the substrate can only distinguish finitely many states. This is not an empirical assumption about physics. It is a logical constraint on any substrate that supports determinate reference. A substrate with infinite local capacity would allow an infinite amount of information to be accessed or distinguished in a single act—which is indistinguishable from having no determinate states at all, because discrimination requires bounded alternatives.*

Definition 3.2 (Maintenance Cost)**.** *Let $W_\psi(\delta, n)$ be the minimum distinguishability work required to maintain ψ's boundary through n successive applications of $\delta \in \Delta(\psi)$. Formally, $W_\psi(\delta, n)$ is the Kolmogorov complexity of the sequence $\langle \psi, \delta(\psi), \delta^2(\psi), \ldots, \delta^n(\psi) \rangle$ relative to the substrate's encoding capacity.*

Remark 3.3. *Kolmogorov complexity is used here as a representative measure of distinguishability work. The argument holds for any monotone complexity measure C satisfying: (i) $C(\langle x_1, \ldots, x_n \rangle) \geq C(x_i)$ for all i, and (ii) C is subadditive up to a constant. These conditions are satisfied by circuit complexity, communication complexity, and all standard Kolmogorov variants. The choice of K is not post hoc; it is the canonical lower bound on any such measure.*

Definition 3.4 (Sustaining Capacity)**.** *Let $\Phi_\psi(n)$ be the maximum work the substrate $\Sigma(\psi)$ can perform across n iterations while maintaining its own integrity. This is bounded by the substrate's finite local capacity: there exists a finite bound $B(\Sigma)$ such that $\Phi_\psi(n) \leq B(\Sigma)$ for all n.*

Theorem 3.5 (Unbounded Cost Eliminates Determinacy)**.** *If $\limsup_{n\to\infty} W_\psi(\delta, n) > B(\Sigma)$ for any $\delta \in \Delta(\psi)$, then ψ does not exist as a determinate form in $\Sigma(\psi)$.*

Proof. Suppose ψ exists as a determinate form in $\Sigma(\psi)$. Then there exists a selective invariant criterion C_ψ distinguishing ψ from at least one alternative, stable under all $\delta \in \Delta(\psi)$. Stability under iteration means $C_\psi(\delta^n(\psi)) = 1$ for all n. For C_ψ to be evaluable at stage n, the substrate must represent the distinction between $\delta^n(\psi)$ and its complement, requiring at least $W_\psi(\delta, n)$ units of substrate capacity. If $\limsup W_\psi(\delta, n)$ exceeds $B(\Sigma)$, then for sufficiently large n the substrate lacks the capacity to maintain the distinction. Therefore C_ψ cannot be stably evaluated for all n, contradicting stability. Hence ψ does not exist as a determinate form in $\Sigma(\psi)$. □

Corollary 3.6 (Finite Representability Constraint)**.** *Any form ψ that exists as a determinate referent in a finite substrate must have a finite upper bound on the Kolmogorov complexity of its iteration sequence under all admissible transformations.*

Remark 3.7. *This explains why mathematical rules persist while mathematical totalities do not. The Peano axioms have finite Kolmogorov complexity; the cost of maintaining the rule is bounded. The completed set of all natural numbers as an extensional totality would require, at each stage of evaluation, the capacity to distinguish infinitely many elements simultaneously—which exceeds any finite substrate's capacity. The rule persists; the infinite extension does not exist as a determinate form in any finite substrate. The framework is neutral on the metaphysical status of infinite substrates; it asserts only that if the substrate is finite, unbounded growth in maintenance cost eliminates determinate existence.*

4 The Eliminative Consequence

Theorem 4.1 (GCC Eliminates Existence)**.** *If ψ contains a globally coupled contradiction, then ψ does not exist as a determinate entity.*

Proof. By Definition 2.4, no stable selective criterion $C_\psi : S \to \{0, 1\}$ exists under $\Delta(\psi)$. By Definition 1.1, conditions (i) and (ii) therefore fail. □

Corollary 4.2 (Violation Is Eliminative)**.** *Forms violating idempotent closure or energetic viability do not exist as determinate entities. The constraint is eliminative, not merely restrictive.*

Proof. Violation of idempotent closure implies the identity condition of ψ drifts or oscillates under $\Delta(\psi)$, which implies no stable C_ψ exists, which by Definition 1.1 eliminates determinate existence. Violation of energetic viability directly fails condition (iii). □

5 Self-Application

A foundational constraint must apply to itself without regress. If it cannot, it is not foundational—it is merely another claim requiring external support.

5.1 The Explicit Calculation

Let ψ^* denote the persistence constraint itself, understood as a form in the sense of Definition 1.1. Its identity condition $\equiv_{\psi^*}$ is: the selective criterion for determinate existence—stable re-identifiability under role-constitutive transformations without globally coupled contradiction and within available sustaining capacity—remains the same criterion under re-application.

5.2 The Admissible Transformations

The role of ψ^* is foundational constraint. The role-constitutive transformations $\Delta(\psi^*)$ are therefore the operations constitutive of that role: evaluation, application to candidate forms, self-application, denial, and formalization. These are fixed prior to evaluation and not chosen post hoc.

5.3 Idempotent Closure: The Formal Check

We verify that $\psi^* \equiv_{\psi^*} \delta^n(\psi^*)$ for all $n \in \mathbb{N}$ and all $\delta \in \Delta(\psi^*)$.

Apply evaluation to ψ^*: the selective criterion C_{ψ^*} selects forms satisfying both conditions and excludes forms violating either. Re-applying C_{ψ^*} to ψ^* returns the same selection. The boundary does not drift. The criterion does not oscillate. Formally:

$$C_{\psi^*}(\delta^n(\psi^*)) = 1 \quad \forall n \in \mathbb{N}, \forall \delta \in \Delta(\psi^*).$$

Idempotent closure holds.

5.4 Energetic Viability: The Explicit Calculation

We compute $W_{\psi^*}(\delta, n)$ and verify $\Phi_{\psi^*}(n) \geq W_{\psi^*}(\delta, n)$ for all n.

The constraint has a finite, fixed description: two conditions, three primitives, one falsification structure. Under any admissible transformation—evaluation, application, self-application, denial, formalization—the constraint returns the same object. The sequence $\langle \psi^*, \delta(\psi^*), \delta^2(\psi^*), \ldots, \delta^n(\psi^*) \rangle$ is therefore the constant sequence $\langle \psi^*, \psi^*, \psi^*, \ldots \rangle$.

The Kolmogorov complexity of a constant sequence is the complexity of a single element plus a finite description of the repetition rule. Both are bounded. Formally:

$$W_{\psi^*}(\delta, n) = K(\psi^*) + O(1) \quad \forall n \in \mathbb{N},$$

where $K(\psi^*)$ denotes the Kolmogorov complexity of the constraint's description and $O(1)$ denotes the constant overhead of the repetition rule. This is bounded above by a finite constant independent of n. Therefore:

$$\Phi_{\psi^*}(n) \geq W_{\psi^*}(\delta, n) = K(\psi^*) + O(1) \quad \forall n \in \mathbb{N}.$$

Energetic viability holds.

Theorem 5.1 (The Constraint Is Self-Sustaining)**.** *The persistence constraint satisfies its own conditions by explicit calculation within the framework.*

Proof. Idempotent closure and energetic viability both hold as demonstrated above. The constraint therefore exists as a determinate entity by its own standard. No regress obtains. □

Remark 5.2. *Any evaluable denial of the constraint presupposes stable targeting of the denied claim across the denial act. Stable targeting requires re-identifiability. Re-identifiability requires the constraint. No evaluable denial therefore succeeds without satisfying the constraint at the level required to aim at it.*[4]

Example 5.3 (Physical Persistence)**.** *Consider a granite formation that has maintained its boundary conditions for one million years. No observer formulated idempotent closure during that period; no selective criterion was explicitly constructed; no formal system tracked its identity. The rock nonetheless satisfies both conditions. Its boundary—the structural constraints that distinguish granite from surrounding material—returns the same form under the admissible transformations constitutive of its role: thermal cycling, pressure variation, chemical exposure. Energetic viability holds because the cost of maintaining its boundary remains within the sustaining capacity of the substrate. The constraint was not applied to the rock. The rock was already satisfying it. This is not a constructed paradox or a limiting case. It is an ordinary physical object obeying a structural requirement that precedes any description of it. The framework is not about how we individuate things. It is about what it takes for there to be something to individuate.*

6 Falsification Conditions

The constraint is refuted by demonstration of any of the following:

(i) a form persisting as determinate while containing GCC at the grounding level with no stable partition witness;

(ii) a form re-identifiable across transformations while lacking any selective criterion;

(iii) a form with unbounded maintenance cost persisting indefinitely on finite sustaining capacity;

(iv) demonstration that this constraint itself contains GCC or violates energetic viability at the level of its own statement.

Absent a demonstrated instance, the constraint stands.

[4] This is the closure of the use-mention gap: a determinate claim cannot be made about the constraint without the constraint already being in place. Denial is not a counterexample; it is a demonstration.

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